25/3/2023 FN

Hall Ticket Number:

MR-21

Code: B313CB6

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Regular Examinations, March-2023

Computer Organization and Architecture

(Common to CSB, CSM & CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Draw the instruction format for memory reference and register reference instructions.	2	1	1	1
1. b)	List four shift micro operations.	2	1	1	1
1. c)	Draw the micro instruction format.	2	1	2	2
1. d)	Differentiate between hardwired and microprogrammed Control unit.	2	4	2	2
1. e)	Compare restoring and non restoring methods.	2	2	5	2
1. f)	What is the range of numbers that can be represented in a fixed point format using 16 bits of a signed integer?	2	2	5	1
1. g)	Compare isolated and memory mapped I/O.	2	2	3	1
1. h)	Define the performance of cache memory in terms of hit ratio with suitable example.	2	1	3	1
1. i)	Draw 8x8 omega switching network.	2	3	4	2
1. j)	List the pipeline conflicts.	2	1	4	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Draw and explain flowchart for instruction cycle.	5	1	1	1
2. b)	Explain about three state bus buffers.	5	2	1	1
	OR				
2. c)	Explain about hardware implementation of 4 bit combinational Arithmetic unit.	5	2	1	1
2. d)	Discuss about some logic micro operations.	5	2	1	1
3. a)	Explain about Micro program sequencer for a control unit.	5	2	2	3
3. b)	Briefly explain the data transfer instructions with an example.	5	2	2	1
	OR				
3. c)	Explain the working of control unit of basic computer system with a suitable diagram.	5	2	2	1
3. d)	Discuss various addressing modes along with examples.	5	2	2	1
4. a)	Perform signed multiplication of -3 and 7 using booth multiplication algorithm. Represent the numbers in 5 bits including sign bit. Represent the step by step multiplication process.	5	3	5	3
4. b)	Explain restoring division algorithm for fixed point numbers with suitable example.	5	2	5	2
	OR				
4. c)	Briefly explain about IEEE floating point representation and perform addition of two floating numbers.	5	2	5	1
4. d)	Draw flow chart for decimal division and explain.	5	1	5	3
5. a)	Briefly explain various modes of data transfer in input output devices	5	2	3	1
5. b)	Draw the block diagram of a asynchronous communication interface and explain.	5	2	3	2
	OR				
5. c)	With neat sketches explain Associative memory.	5	3	3	1
5. d)	Illustrate Daisy chain priority interrupt with diagram.	5	3	3	1
6. a)	Define Parallel processing and what are the different types of parallel processing? How parallelism can be achieved in uni-processor system?	5	1	4	5
6. b)	Explain about parallel arbitration logic.	5	2	4	5
	OR				
6. c)	Explain about memory interleaving concept in vector processing.	5	2	4	4
6. d)	Explain about four segment CPU pipeline with the help of flow chart.	5	2	4	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome